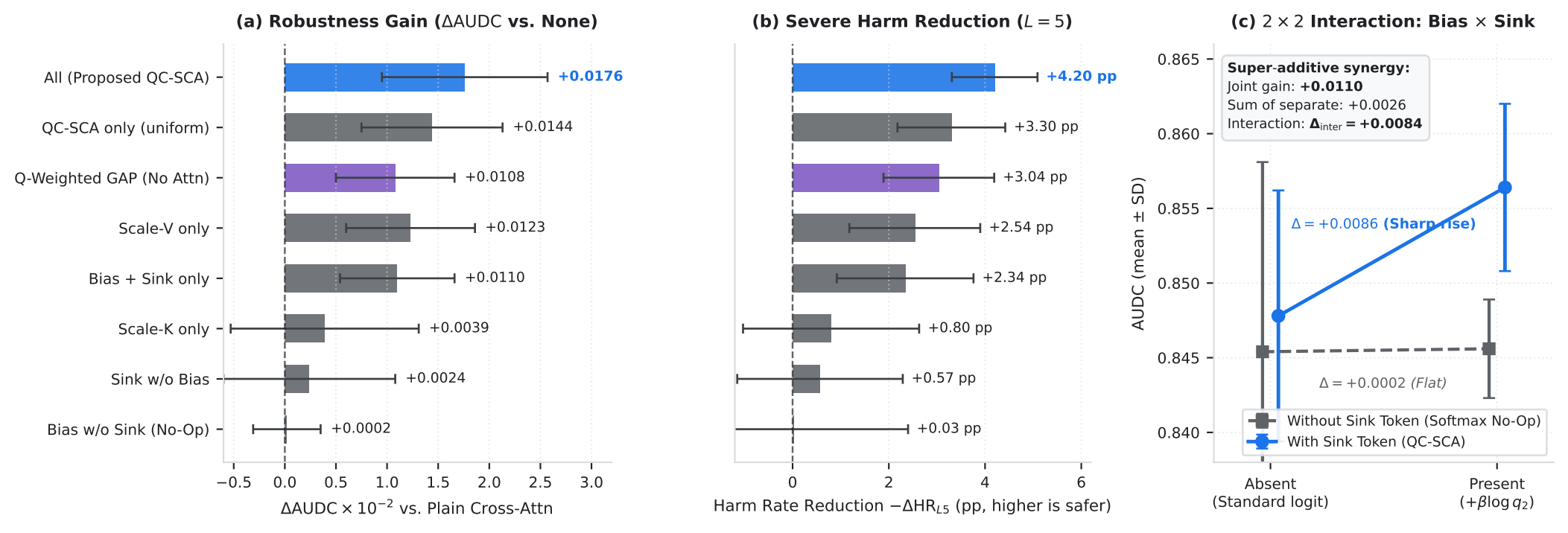
(a) Robustness Gain (ΔAUDC vs. None)
All (Proposed QC-SCA)
QC-SCA only (uniform)
Q-Weighted GAP (No Attn)
Scale-V only
Bias + Sink only
Scale-K only
Sink w/o Bias
Bias w/o Sink (No-Op)
+0.0176
+0.0144
+0.0108
+0.0123
+0.0110
+0.0039
+0.0024
+0.0002
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
ΔAUDC × 10−2 vs. Plain Cross-Attn
(b) Severe Harm Reduction (L = 5)
+4.20 pp
+3.30 pp
+3.04 pp
+2.54 pp
+2.34 pp
+0.80 pp
+0.57 pp
+0.03 pp
0
2
4
6
Harm Rate Reduction −ΔHRL5 (pp, higher is safer)
(c) 2 × 2 Interaction: Bias × Sink
Super-additive synergy:
Joint gain: +0.0110
Sum of separate: +0.0026
Interaction: Δinter = +0.0084
Δ = +0.0086 (Sharp rise)
Δ = +0.0002 (Flat)
Without Sink Token (Softmax No-Op)
With Sink Token (QC-SCA)
AUDC (mean ± SD)
0.865
0.860
0.855
0.850
0.845
0.840
Absent
(Standard logit)
Present
(+β log q2)